

Blaine Heffron

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Summary

Software engineer with 7+ years of professional development experience and 12+ years writing data-heavy software across product engineering, research, and consulting. My work has centered on Java, C#, Rust, TypeScript/JavaScript, and Python across AWS-backed services, developer tooling, PostgreSQL data systems, authentication flows, and API integrations. Physics PhD with a strong record of documentation, debugging, software architecture, and shipping reliable systems in technically messy environments.

Experience

Software Engineer / Smart Contract Tooling Contractor

2024 – 2025

AhaLabs / Stellar Ecosystem

Remote

- Built Rust and TypeScript tooling for Stellar smart contract development, including work on Scaffold Stellar build and watch workflows, generated TypeScript bindings, and registry-based deployment tooling.
- Developed EquitX contract and backend support systems, including sync and indexing services for on-chain CDPs, staking positions, liquidity pools, price feeds, and automated liquidations using TypeScript, Express, and PostgreSQL.
- Won a \$60k Stellar Community Fund grant for *soroban-sdk-tools*, an open-source Rust crate that cuts down smart contract boilerplate around storage, error handling, imports, and auth testing.
- Contributed to ecosystem tooling used by contract teams, with an emphasis on clearer interfaces and fewer manual deployment steps.

Software Engineer (Contract)

Oct 2024 – Aug 2025

Breeze Homes

Remote

- Built and maintained a TypeScript-based web platform for customer onboarding, project tracking, and contractor workflows using React, Vite, Material UI, and Node/AWS serverless services.
- Designed and implemented cloud-backed APIs on AWS Lambda, API Gateway, S3, and SNS with PostgreSQL and TypeORM, including project and sub-project CRUD, floor plan handling, photo storage, search, and pagination.
- Integrated secure authentication and authorization with SuperTokens, API keys, session refresh, and role or ownership checks for admin, contractor, and end-user access paths.
- Worked through production deployment, CORS, and data consistency issues across environments, and supported HubSpot and Pinterest integrations for downstream business workflows.

Independent Consultant / Software Developer

Feb 2023 – Present

Self-Employed

Charleston, SC

- Deliver full-stack applications, workflow automation, and internal tools in Python, TypeScript, Rust, and C++ for clients with messy operational problems and evolving requirements.
- Build and maintain web applications, data pipelines, API integrations, and internal developer tools, with an emphasis on maintainability, debugging, and clear handoff documentation.
- Lead developer for Songcraft, an educational music game that helped secure more than \$40k in funding.

Technical Support Engineer

Mar 2018 – Aug 2019

CData Software

Remote

- Supported enterprise Java and C# data connectivity products and customer integrations across databases, APIs, and application environments, reproducing and resolving complex technical issues.
- Did development work to automate parts of the driver creation process, including an internal tool that generated integrations from OpenAPI specifications and reduced manual setup work.
- Worked directly with customers and internal engineering teams to diagnose failures, document fixes, and unblock high-priority deployments.

Graduate Research Assistant

University of Tennessee, Department of Physics

Aug 2019 – Jan 2023

Knoxville, TN

- Developed and maintained analysis software in Python and C++ for the PROSPECT neutrino experiment, including machine learning workflows, data validation, and research tooling used in a multi-institution collaboration.
- Planned and executed HFIR radiation field measurements with HPGe instrumentation, then analyzed large datasets to improve understanding of reactor background sources and detector constraints.
- Mentored undergraduate and graduate researchers on software workflows, debugging, analysis methods, and technical communication.

Graduate Research Assistant

University of Tennessee, Department of Physics

Aug 2014 – Dec 2016

Knoxville, TN

- Built simulation and analysis tooling for reactor and cosmogenic background studies supporting the PROSPECT detector program at Oak Ridge National Laboratory.
- Produced documented research software and technical analysis for detector characterization, shielding studies, and experimental planning.

Research Assistant

UNC Chapel Hill, Environmental Sciences and Engineering

Nov 2012 – Aug 2014

Chapel Hill, NC

- Built Python modeling tools for sunlight flux and smog chamber simulation, including nonlinear differential equation models for aerosol formation and dynamics.

Education

PhD, Physics

University of Tennessee

2023

Knoxville, TN

MS, Physics

University of Tennessee

2017

Knoxville, TN

BS, Physics

University of North Carolina at Chapel Hill

2012

Chapel Hill, NC

Technical Skills

Languages: Java, C#, Rust, TypeScript, JavaScript, Python, C++, SQL

Cloud and Data: AWS Lambda, API Gateway, S3, SNS, PostgreSQL, SQLite, TypeORM, REST APIs, serverless deployments, event indexing

Security and Architecture: Authentication and session management, role-based access control, API key protection, secure integration patterns, software design, software architecture, microservices, debugging, scalable service design

Frontend and Tooling: React, Vite, Material UI, Git, Postman, React Testing Library, Linux, Cargo, WASM tooling

Process: Agile development, code review, technical documentation, cross-team collaboration, issue triage

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